# Accenture Japan — IIT Dhanbad

Online Assessment Pattern & Coding Questions

## 1. Overall OA Pattern

| Section | Questions | Time |
|---|---|---|
| MCQs | 20 | 20 minutes |
| Coding / DSA | 1 | 60 minutes |
| Total | 21 | 80 minutes |

The assessment consisted of **20 MCQs** followed by **1 coding question**. The MCQ section had a time limit of **20 minutes**, while the coding section had a time limit of **60 minutes**.

## 2. MCQ Section — 20 Questions

The MCQs covered a variety of topics, including:

- **LLM / Generative AI**
- **Aptitude**
- **Pseudocode / Output-based questions**
- **SQL**
- **Computer Science Fundamentals** — mainly Operating Systems and Cloud Computing

The OS and Cloud Computing questions were reported to be **relatively easy**.

## 3. Coding / DSA Section — 60 Minutes

There was **1 coding question**, with **60 minutes** allocated to the coding section. **All programming languages were allowed.**

### DSA Question — Set 1: Morse Code Dictionary

A Morse-code mapping table was provided, where each English alphabet character corresponds to a Morse-code representation using combinations of **.** and **-**.

Given a string written in **Morse code**, decode it and output a corresponding English word.

The decoded word must be present in the provided **dictionary.txt** file.

**Dictionary constraint:**

- The output word must be present in **dictionary.txt**.
- The word's Morse-code representation must match the given input.
- If multiple valid dictionary words can be generated, **any one valid word** may be printed.

**Input:** A Morse-code string is provided as input, along with the Morse-code mapping table for the alphabet.

**Output:** Print any valid word from **dictionary.txt** whose Morse-code representation matches the given input.

### DSA Question — Set 2: Minimum Dice Rolls to Reach the End

You are given an array **A** of non-negative integers of size **N**. You start at **index 0** and need to reach an index **greater than or equal to N**.

A special die is available. You can choose **any number from 1 to 6** whenever you roll the die.

**At any index, there are two possible actions:**

- **Roll the die:** choose any value from 1 to 6 and jump forward by that many positions. This counts as **one dice roll**.
- **Use the value at the current index:** jump forward by exactly **A[i]** positions. This action does **not** require a dice roll.

The objective is to reach an index **>= N** using the **minimum possible number of dice rolls**.

**Example**

```
N = 10

A = [0, 0, 0, 0, 0, 1, 1, 1, 1]

Output: 1
```

**Reference:** The core jump/reach structure is similar to **LeetCode 45 — Jump Game II**, but this problem introduces an additional special-die operation and free jumps based on the current array value.

# 4. OA Summary

| Section | Questions | Time | Topics |
|---|---|---|---|
| MCQs | 20 | 20 min | LLM, Aptitude, Pseudocode, SQL, OS, Cloud Computing |
| Coding | 1 | 60 min | Morse Code Dictionary / Minimum Dice Rolls |
| Total | 21 | 80 min | — |